PNP–BV Inverse Problem: Week of March 4, 2026
Inference Accuracy Exploration

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1 Problem Statement

The multi-observable master inference pipeline (v3–v9) simultaneously recovers all four Butler–Volmer kinetic parameters ($k_{0,1}$, $k_{0,2}$, α_1, α_2) for a 4-species PNP–BV electrochemical system. However, a persistent $k_{0,1}$–$k_{0,2}$ **accuracy trade-off** prevents any version from achieving $< 10\%$ error on all four parameters simultaneously. The root cause is a 24× scale disparity between the two exchange rate constants ($k_{0,1} = 2.4 \times 10^{-8}$ m/s, $k_{0,2} = 1.0 \times 10^{-9}$ m/s), which causes the total current observable to be dominated by R1 and creates gradient competition in the least-squares objective when both observables are equally weighted.

Prior results (Table 1) showed complementary strengths: v7 (PDE-based, parallel) achieved $k_{0,2}$ error of 2.6% but $k_{0,1}$ error of 10.9%; v8 (surrogate, 181 samples) achieved $k_{0,1}$ error of 7.3% but $k_{0,2}$ error of 9.6%. No version had crossed the $< 10\%$ threshold on all parameters—until **v11 (Strategy 7)**, which combines surrogate warm-starting with PDE refinement to achieve 9.43% max error with all four parameters below 10%.

Table 1: Best parameter recovery errors (%) by pipeline version. All use 2% Gaussian noise on synthetic targets.

Version	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
v7 (PDE)	10.9	2.6	5.6	8.8	10.9
v8 (Surr., 181)	7.3	9.6	4.5	4.6	9.6
v9 (Surr., 445)	8.8	7.6	4.8	6.4	8.8
v11 (Surr.+PDE)	**9.4**	**0.8**	**5.1**	**7.8**	**9.4**

2 Exploration Strategies

Seven strategies were proposed; all seven were investigated and completed this week.

1. **Adaptive Observable Weighting.** Swept $w_2 \in \{0.1, 0.25, 0.5, 1, 2, 5, 10, 20\}$ on the v9 surrogate. **Result:** $w_2 = 1.0$ is already minimax-optimal (max err = 8.76%). **Complete.**

2. **Block Coordinate Descent (BCD).** Alternated R1/R2 parameter blocks (30 iterations, full voltage grid). **Result:** max err = 22.10% ($k_{0,2}$ degraded due to flat valley on full grid). **Complete.**

3. **Multi-Start LHS Grid Search.** 20K LHS points + 20 polished candidates. **Result:** all converge to the same 8.76% optimum—confirmed global minimum. **Complete.**

4. **Per-Observable Cascade.** Use $w_2 = 0.5$ to lock in $k_{0,1}$, then fix R1 params and optimize $k_{0,2}$ in 2D search. **Result:** 3/4 params excellent, $k_{0,2} = 16.60\%$; polish erases gains. **Complete.**

5. **Fixed PDE Refinement.** Re-enable PDE with tight bounds, Tikhonov regularization, and early stopping. **Result:** max err = 8.73%; PDE no longer regresses. **Complete.**

6. **Cascade + PDE Hybrid.** Cascade for 3/4 params, then PDE refinement on $k_{0,2}$. Four sub-iterations: (6a) original 1D, 14.91%; (6b) multi-observable, 12.51%; (6c) asymmetric regularization, 12.51%; (6d) free joint PDE, 13.60% (Pareto tradeoff). **Root cause:** tight 4-parameter coupling biases $k_{0,2}$ by 12% when 3 params are fixed; freeing all 4 reveals a fundamental $k_{0,1}$–$k_{0,2}$ anticorrelation.

7. **Surrogate-Warm-Started PDE Pipeline (v11).** Replace v7's expensive cold-start PDE phases with the v9 surrogate for warm-starting, then run v7's proven PDE protocol with wide bounds, no regularization, and the full cathodic grid. **Result:** max err = 9.43%; all four parameters < 10% simultaneously for the first time. $k_{0,2} = 0.82\%$ (best ever). **Complete.**

3 Strategy 1: Adaptive Observable Weighting

Swept $w_2 \in \{0.1, 0.25, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$ on the v9 surrogate (Phase 2 joint optimization, shallow cathodic grid).

Table 2: Weight sweep results. Bold row is the minimax optimum.

w_2	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
0.5	0.5	17.6	5.1	2.8	17.6
1.0	**8.8**	**7.6**	**4.8**	**6.4**	**8.8**
2.0	19.6	3.5	4.6	7.4	19.6

The trade-off is monotonic: lowering w_2 improves $k_{0,1}$ at the cost of $k_{0,2}$ (and vice versa). $w_2 = 1.0$ is already the minimax optimum at 8.76%. **Simple reweighting cannot break the trade-off.**

4 Strategy 2: Block Coordinate Descent

BCD alternates between optimizing R1 and R2 parameter blocks (30 iterations, full voltage grid). Best result: $k_{0,1} = 7.39\%$, $k_{0,2} = 22.10\%$, $\alpha_1 = 3.03\%$, $\alpha_2 = 4.18\%$ (max = 22.10%). Alpha recovery was excellent, but $k_{0,2}$ degraded 3× vs. baseline because BCD used the full voltage grid where the plateau regime creates a flat valley in the $k_{0,2}$ direction. **BCD alone does not help.**

5 Strategy 3: Multi-Start LHS Grid Search

20,000 Latin Hypercube samples spanning the full 4D parameter space, with the top 20 candidates polished by L-BFGS-B. **All 20 converged to the same optimum** (max err = 8.76%). This confirms 8.76% is the **global optimum** of the surrogate landscape. No optimizer can beat it.

6 Root Cause: Surrogate Fidelity

The bottleneck is the peroxide current surrogate, which has NRMSE = 11.43% (vs. 0.42% for current density). Three factors drive this:

- **Cancellation error:** $I_{\text{pxd}} = -(R_1 - R_2)$ involves subtracting similar quantities ($R_1 \gg R_2$), amplifying relative interpolation errors by $\sim$24$\times$.
- **Training sparsity:** Zero of 445 LHS samples fall within 0.1 log-decades of both true k_0 values simultaneously.
- **PDE refinement overfitting:** The exact PDE solver fits the 2% noise realization, decreasing the objective while increasing parameter error.

7 Strategy 4: Per-Observable Cascade

Used $w_2 = 0.5$ (CD-dominant) to lock in $k_{0,1}$, α_1, α_2, then fixed those parameters and re-optimized $[k_{0,2}, \alpha_2]$ in a reduced 2D search with $w_2 = 2.0$.

Table 3: Per-observable cascade results.

Stage	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
v9 baseline	8.8	7.6	4.8	6.4	8.8
Cascade (no polish)	0.5	16.6	1.9	2.4	16.6
Cascade + polish	8.8	7.6	4.8	6.4	8.8

The cascade achieves excellent accuracy for 3/4 parameters ($k_{0,1} = 0.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.38\%$), but $k_{0,2}$ remains the sole bottleneck at 16.60%. Joint polish erases all cascade gains, converging back to the v9 global optimum—**the surrogate has a single basin**.

8 Strategy 5: Fixed PDE Refinement

Re-enabled PDE refinement with three safeguards: tight bounds ($\pm$0.3 log-decades), Tikhonov regularization ($\lambda = 0.1$), and early stopping (10 iterations).

Table 4: Fixed PDE refinement results (best: $\lambda = 0.1$).

Method	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
v9 baseline	8.8	7.6	4.8	6.4	8.8
v9 + PDE (old, no guard)	16.8	—	—	—	16.8
v9 + PDE ($\lambda = 0.1$)	8.7	7.5	4.9	6.2	8.7

The improvement is marginal (8.73% vs. 8.76%), but critically **PDE no longer regresses** (was 16.78% without safeguards). The three fixes work as defense-in-depth.

9 Strategy 6: Cascade + PDE Hybrid

Combined the cascade (S4) to lock in 3/4 parameters with PDE refinement on $k_{0,2}$. Four progressively refined sub-iterations were investigated.

9.1 6a: Original Hybrid (shallow voltages, $w = 1.0$, `minimize_scalar`)

Table 5: Sub-iteration 6a: original cascade + PDE hybrid.

Stage	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
v9 baseline	8.8	7.6	4.8	6.4	8.8
Phase 1 (cascade)	0.5	16.6	1.9	2.1	16.6
Phase 2 (PDE $k_{0,2}$)	0.5	14.9	1.9	2.4	14.9

PDE reduces $k_{0,2}$ from 16.60% to 14.91%, but the objective varies by only $\sim 5 \times 10^{-8}$ across the $k_{0,2}$ range—**extremely flat** at shallow voltages.

9.2 6b: Multi-Observable PDE (full cathodic, $w = 5.0$, L-BFGS-B)

Extended to the full cathodic grid and switched to gradient-based L-BFGS-B.

Table 6: Sub-iteration 6b: multi-observable PDE, full cathodic range.

Stage	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
6a result	0.5	14.9	1.9	2.4	14.9
6b (full cat., w=5, BFGS)	0.5	12.5	1.9	2.4	12.5

Improvement from 14.91% to 12.51% due to full cathodic range + gradient use. A weight sweep ($w_2 \in \{1, 2, 5, 10, 20\}$) showed **all weights converge to the same** $k_{0,2} \approx 12.5\%$—with one free parameter, the 1D minimum is unique regardless of weight.

9.3 6c: Asymmetric Regularization

Tested per-parameter Tikhonov penalties ($\lambda_{k_{0,1}} = 5.0$, $\lambda_{k_{0,2}} = 0.1$, $\lambda_{\alpha_1} = 5.0$, $\lambda_{\alpha_2} = 5.0$) to anchor 3 params while freeing $k_{0,2}$.

Table 7: Sub-iteration 6c: asymmetric regularization and lambda sweep.

Configuration	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
6b (1D, w=5)	0.5	12.5	1.9	2.4	12.5
6c ($\lambda_{k_{0,2}}$=0.1)	0.5	12.5	1.9	2.3	12.5
6c ($\lambda_{k_{0,2}}$=0.0)	0.5	12.5	1.9	2.3	12.5
6c ($\lambda_{k_{0,2}}$=0.01)	0.5	12.5	1.9	2.3	12.5

All lambda values converge to the same 12.51%. **Root cause identified:** the cascade's small errors in 3 params (0.51%, 1.92%, 2.38%) bias the PDE-optimal $k_{0,2}$ by $\sim$12% because the four parameters are *tightly coupled* in the PDE objective. v7 achieved $k_{0,2} = 2.58\%$ because all 4 moved freely via joint Hessian coupling.

9.4 6d: Free Joint PDE Mode

The `--joint-mode free` option uses the cascade result as a warm start for joint 4-parameter PDE optimization with moderate bounds ($k_{0,1}$ factor = 3.0, $k_{0,2}$ factor = 5.0, α margin = 0.08) and configurable regularization.

Table 8: Sub-iteration 6d: free joint PDE mode configurations. Phase 3 regressed in all 4 cases; the pipeline kept Phase 2 (12.51%) as the final answer.

Config	λ	$\lambda_{k_{0,2}}$	Iters	w_2	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
1: No reg	0.0	0.0	15	5.0	19.7	11.5	7.6	6.5	19.7
2: No reg	0.0	0.0	25	5.0	37.8	5.4	14.0	15.1	37.8
3: Sym reg	0.01	0.01	15	5.0	4.9	13.6	1.4	0.3	13.6
4: No reg	0.0	0.0	15	1.0	6.8	18.4	4.2	2.8	18.4

Key findings:

- **Config 2** achieved $k_{0,2} = 5.40\%$ (near the $< 5\%$ target) but $k_{0,1}$ exploded to 37.82%—the optimizer trades $k_{0,1}$ accuracy for $k_{0,2}$.
- **Config 3** achieved $k_{0,1} = 4.90\%$, $\alpha_1 = 1.43\%$, $\alpha_2 = 0.29\%$ (all below 5%) but $k_{0,2} = 13.60\%$.
- The four desired sub-5% values exist *across* different configs but **not simultaneously** in any single config.
- $k_{0,1}$ and $k_{0,2}$ are **anticorrelated** in the PDE objective landscape—a fundamental Pareto tradeoff that cannot be resolved by optimizer tuning alone.

10 Strategy 7: Surrogate-Warm-Started PDE Pipeline (v11)

The key insight from Strategies 1–6 is that the surrogate landscape has a single basin (8.76% max error, confirmed global by Strategy 3), and PDE refinement with tight bounds merely stays near that basin (Strategy 5) or regresses (v10). Meanwhile, v7's *wide-bound, full-cathodic PDE protocol* achieved $k_{0,2} = 2.6\%$—the best ever—but required 345.0 s of cold-start PDE computation. Strategy 7 combines both: use the surrogate for a fast warm start, then run v7's proven PDE protocol.

10.1 Architecture: 4-Phase Pipeline

1. **Phase 1: Surrogate alpha-only** (< 0.1 s). Fix k_0 at initial guess [0.005, 0.0005]; optimize $[\alpha_1, \alpha_2]$ via `AlphaOnlySurrogateObjective` on the full 22-point voltage grid. L-BFGS-B, 60 maxiter.
2. **Phase 2: Surrogate joint 4-param** (< 0.1 s). Cold k_0 start, alpha warm from Phase 1. `SubsetSurrogateObjective` on the shallow cathodic subset (10 pts). L-BFGS-B, 60 maxiter, bounds from surrogate training range.
3. **Phase 3: PDE joint on shallow cathodic** ($\sim$120 s). Warm start from Phase 2 surrogate. 10-point grid ($\hat{\eta} \in [-1, -13]$). L-BFGS-B, maxiter = 30, `ftol` $= 10^{-8}$, `gtol` $= 5 \times 10^{-6}$. **Wide bounds:** $k_0 \in [10^{-8}, 100]$ (log-space), $\alpha \in [0.05, 0.95]$. **No regularization.** Shared parallel pool (9 workers).

4. **Phase 4: PDE joint on full cathodic** (∼120 s). Warm start from Phase 3. 15-point grid ($\hat{\eta} \in [-1, -46.5]$). Same optimizer, bounds, and pool as Phase 3 with maxiter = 25.

The best-of selection compares Phase 3 and Phase 4 by max relative error and keeps the better result (regression guard from v7).

10.2 Results: 2% Noise (Default)

Table 9: v11 phase-by-phase results with 2% Gaussian noise on synthetic targets.

Phase	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Time
P1 (surr. α-only)	295.8	850.0	43.8	64.8	850.0	< 0.1 s
P2 (surr. joint)	8.8	7.6	4.8	6.4	8.8	< 0.1 s
P3 (PDE shallow)	16.5	11.8	8.6	7.5	16.5	239 s
P4 (PDE full cat.)	**9.4**	**0.8**	**5.1**	**7.8**	**9.4**	365 s

Phase 4 is selected as the best result (max err = 9.43% vs. Phase 3's 16.46%). The full cathodic grid dramatically improves $k_{0,2}$: from 11.8% (shallow) to **0.82%** (full cathodic)—confirming that the deep-voltage R2 sensitivity identified in the root-cause analysis (§6) is essential for $k_{0,2}$ recovery.

10.3 Noise-Free Diagnostic (0% Noise)

To separate surrogate error from noise-induced error, v11 was run with 0% noise on the synthetic targets.

Table 10: v11 noise-free diagnostic. Surrogate error alone produces ∼1% parameter error; PDE refinement reduces it further.

Phase	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Time
P2 (surr. joint)	0.43	1.02	0.28	0.44	1.02	< 0.1 s
P3 (PDE shallow)	0.47	1.08	0.11	0.20	1.08	106 s
P4 (PDE full cat.)	**0.45**	**1.07**	**0.07**	**0.13**	**1.07**	137 s

Without noise, the surrogate alone achieves ∼1% max error, and PDE refinement reduces it to 1.07% (all parameters $< 1.1\%$). This confirms that the surrogate warm start is accurate and that **the 2% noise is the dominant source of residual error** in the noisy case, not the surrogate or the PDE solver.

10.4 Timing Comparison

The surrogate warm start eliminates v7's two cold-start PDE phases (∼100 s saved), yielding a **30% reduction in PDE wall time** (243 s vs. 345 s).

11 Summary

Key conclusions:

Table 11: Wall-time comparison: v7 (pure PDE) vs. v11 (surrogate + PDE). Target generation ($\sim$76 s) excluded from optimization time.

Pipeline	PDE time	Total time	Speedup
v7 (cold start)	$\sim$345 s	$\sim$415 s	—
v11 (surrogate + PDE)	$\sim$243 s	$\sim$319 s	1.4×

Table 12: Strategy comparison: per-parameter errors and overall result. Strategy 6 expanded to show all sub-iterations. Strategy 7 (v11) achieves $< 10\%$ on all four parameters.

S#	Strategy	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Status
1	Weight sweep	8.8	7.6	4.8	6.4	8.8	Default optimal
2	BCD	7.4	22.1	3.0	4.2	22.1	Worse
3	Multi-start	8.8	7.6	4.8	6.4	8.8	Global opt. confirmed
4	Cascade	0.5	16.6	1.9	2.4	16.6	Best 3/4 params
5	Fixed PDE	8.7	7.5	4.9	6.2	8.7	Stops regression
Strategy 6: Cascade + PDE Hybrid (sub-iterations)							
6a	Original (shallow, w=1)	0.5	14.9	1.9	2.4	14.9	PDE flat at shallow V
6b	Multi-obs. (full cat., w=5)	0.5	12.5	1.9	2.4	12.5	Weight-invariant
6c	Asym. reg. (λ_{k_2}=0.1)	0.5	12.5	1.9	2.3	12.5	Coupling root cause
6d	Free joint PDE (best)	4.9	13.6	1.4	0.3	13.6	Pareto tradeoff
Strategy 7: Surrogate-Warm-Started PDE (v11)							
7	**Surr.+PDE (v11, P4)**	**9.43**	**0.82**	**5.13**	**7.85**	**9.43**	**All** $< 10\%$
7	Surr.+PDE (v11, P3)	16.5	11.8	8.6	7.5	16.5	Shallow only
7	Surr. only (v11 P2)	8.8	7.6	4.8	6.4	8.8	= v9 baseline

- **Strategy 7 (v11) achieves** $< 10\%$ **on all four parameters for the first time** (max err $= 9.43\%$). The key was combining the v9 surrogate for warm-starting with v7's wide-bound, full-cathodic PDE protocol—not optimizer tuning on a fixed surrogate (Strategies 1–6).
- **The surrogate-only landscape has a single basin** at 8.76% max error (confirmed global by 20K-point multi-start, Strategy 3). Optimizer tuning alone (Strategies 1–6) cannot beat this.
- **PDE refinement with wide bounds breaks the surrogate ceiling.** Phase 4's full cathodic grid reduces $k_{0,2}$ from 7.57% (surrogate) to **0.82%** (PDE)—the best $k_{0,2}$ accuracy achieved by any version, beating even v7's 2.6%.
- **The** $k_{0,1}$–$k_{0,2}$ **trade-off persists** but is less severe with v11's warm start. Compared to v7, all four errors improve: $k_{0,1}$: 10.9%→9.4%, $k_{0,2}$: 2.6%→0.8%, α_1: 5.6%→5.1%, α_2: 8.8%→7.8%.
- **Noise is the dominant error source**, not surrogate fidelity. The noise-free diagnostic achieves 1.07% max error (all params $< 1.1\%$), confirming the surrogate warm start is accurate and PDE refinement converges to near-truth without noise.
- **30% PDE wall-time reduction** (243 s vs. v7's 345 s) from eliminating two cold-start PDE phases via surrogate warm-starting.
- **Surrogate PC NRMSE = 11.43%** remains a fidelity concern (27× worse than the CD surrogate at 0.42%), but the PDE refinement successfully compensates.
- **The cascade approach** ($k_{0,1} = 0.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.38\%$) **remains valuable** when $k_{0,1}/\alpha$ accuracy matters more than $k_{0,2}$.

12 Next Steps

1. **Push** $k_{0,1}$ **below 5% (primary path):** v11 achieves $k_{0,1} = 9.4\%$—the largest remaining error. Two levers: (a) adaptive surrogate retraining with ∼500 focused PDE samples near the v11 best estimate to tighten the warm start; (b) increasing Phase 4 `maxiter` from 25 to 50–80 to allow the PDE optimizer more freedom to refine $k_{0,1}$.
2. **Noise robustness study:** v11 achieves 1.07% max error at 0% noise but 9.43% at 2% noise—a 9× degradation. Investigate noise-aware techniques (ensemble averaging, Bayesian regularization) to close this gap.
3. **Surrogate-predicted steady states (future work):** If the surrogate could predict full spatial concentration fields (e.g., via POD+RBF), Phase 3's first sequential voltage sweep could be eliminated, cutting PDE time roughly in half. Requires a fundamentally different surrogate architecture.